YANSHAN UNIVERSITY

School of Electrical Engineering

DEEP LEARNING-BASED THREE-PHASE VOLTAGE PARAMETER ESTIMATION UNDER DISTURBANCES: A STUDY OF SAMPLING FREQUENCY AND OBSERVATION WINDOW

A Research Report

BY

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ABSTRACT

Accurate estimation of three-phase voltage parameters is essential for power system monitoring under disturbances such as harmonics, noise, DC offset, and frequency deviation. This study investigates deep supervised learning for estimating fundamental voltage amplitudes, phase angles, and frequency, with the primary objective of identifying configurations that reduce conventional requirements of 10 kHz sampling and 20 ms observation windows while maintaining estimation accuracy.

A feedforward neural network with seven regression outputs was evaluated using simulated voltage signals through four experiments examining dataset size, sampling frequency, observation window, and their combined effects. Performance was rigorously assessed using Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and training time. The results confirm that a full 20 ms fundamental-cycle window is critical for minimizing both average errors and outlier variance (RMSE). While 10 kHz sampling with 20,000 datasets yields the lowest absolute error, a "Goldilocks" analysis reveals that 5 kHz sampling with a 20 ms window and 10,000 datasets provides the optimal balance—halving the input dimension and training time with only marginal accuracy loss. The findings offer validated, application-specific configurations (ranging from high-precision offline processing to low-latency monitoring) that effectively reduce data acquisition requirements. Further validation using real power-system data and edge-device inference latency testing is recommended.

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1. INTRODUCTION

Three-phase voltage parameter estimation is essential for power-system monitoring and control, especially under voltage imbalance and waveform disturbances. Deep learning offers a data-driven approach to estimating voltage amplitudes, phase angles, and fundamental frequency directly from sampled voltage signals.

2. PROBLEM STATEMENT

Disturbances such as harmonics, measurement noise, DC offset, and frequency variations can affect estimation accuracy. Furthermore, sampling frequency, observation-window duration, and training dataset size influence the information available to the neural network and its computational requirements. Determining an effective balance among these factors is the central problem addressed in this study.

This report consists of five chapters covering the introduction, problem statement, methodology, results and discussion, and conclusion. The appendices provide detailed experimental results, supplementary figures, tables, and MATLAB code.

3 RESEARCH METHOD

3.1 Mathematical Model of the Three-Phase Voltage

In a power system, the three-phase voltages ideally have equal amplitudes, the same frequency, and a phase displacement of 120°. In practice, disturbances such as harmonics, noise, DC offset, and voltage imbalance can cause deviations from these ideal conditions. This study uses a mathematical model to simulate these disturbances and investigate how accurately a neural network can estimate the fundamental voltage parameters.

$$v_p(t) = A_p \sin(2\pi ft+\phi_p) + A_{5,p} \sin(2\pi 5ft+\phi_{5,p}) + A_{7,p} \sin(2\pi 7ft+\phi_{7,p}) + D_p+n_p(t)$$

where (A_p), (ϕ_p), and (f) represent the fundamental peak amplitude, phase angle, and frequency, respectively. The terms $(A_{5,p})$ and$(A_{7,p})$represent the fifth- and seventh-harmonic amplitudes,(D_p) is the DC offset, and $(n_p(t))$ represents additive noise.

The fundamental frequency(f) is common to all three phases, while their fundamental amplitudes and phase angles may differ. The fifth and seventh harmonics are included to reproduce waveform distortion, while the DC offset and additive noise represent additional deviations from an ideal sinusoidal waveform. The equation above describes the signal-generation model used in the simulation.

3.1.2 Sampling and Observation Window

Continuous-time phase voltages are discretized before entering the neural network. The number of samples per phase is:

$$N = round\,(f_s T_{\text{obs}}),$$

where N is the number of samples per phase,(f_s) is the sampling frequency and (T_{obs}) is the observation-window duration. The sampled voltage is expressed as

$$v_p[k] = v_p\left(\frac{k}{f_s}\right), \quad k = 0,1, \ldots, N-1.$$

At the nominal 50 Hz grid frequency,

$$T_0 = \frac{1}{50} = 0.02s = 20ms,$$

So a 20 ms window equals one fundamental cycle. Shorter windows contain only part of a cycle and provide less waveform information to the estimator.

The three phase signals are concatenated as

$$[\mathbf{x} = v_a[0], \ldots, v_a[N-1], v_b[0], \ldots v_b[N-1], v_c[0], \ldots, v_c[N-1]$$

The resulting input dimension is $\mathbf{3}\boldsymbol{N}$. Therefore, the input size changes with both sampling frequency and observation-window duration.

3.1.3 Disturbance Conditions and Simulation Parameters

The datasets contain voltage signals with varying fundamental amplitudes, frequency deviations, phase-angle deviations, harmonic distortion, DC offsets, and additive noise. The severity of these disturbances is varied to expose the network to a range of operating conditions.

Table 3.1: Principal simulation parameters and their purposes

Parameter	Values or range
Nominal fundamental frequency	50 Hz
Fundamental frequency range	49.5–50.5 Hz in the combined experiment
Nominal phase voltage	220 V RMS
Nominal peak voltage	Approximately 311.13 V
Sampling frequency	1, 2, 5 and 10 kHz
Observation window	2, 5, 10, 15 and 20 ms
Dataset size	1,000–20,000 examples, depending on the experiment
Harmonic orders	Fifth and seventh
Harmonic amplitude	0–5% for moderate conditions; 5–10% for severe conditions in the combined experiment
Signal-to-noise ratio	30–40 dB for moderate conditions; 20–30 dB for severe conditions in the combined experiment
DC offset	Up to 2% for moderate conditions and 5% for severe conditions in the combined experiment
Fundamental amplitude variation	Approximately 90–110% of nominal peak amplitude in the combined experiment
Phase-angle deviation	Within approximately ±5° of the nominal phase displacement in the combined experiment
Training, validation and internal test proportions	70%, 15% and 15%

The parameters in Table 3.1 were selected using established power-system principles and practical considerations, while some numerical ranges were introduced as simulation assumptions. Power frequency, voltage magnitude, voltage unbalance, and harmonic distortion are recognised power-quality parameters in IEC 61000-4-30 and IEEE 1159-2019 [1], [2]. These references support the choice of phenomena to investigate, but they do not prescribe all the numerical ranges used in this study.

The nominal frequency was set to 50 Hz, and the nominal phase voltage was set to 220 V RMS, corresponding to approximately 311.13 V peak. The frequency range of 49.5 – 50.5 Hz and the amplitude variation of 90 – 110% of nominal peak voltage were selected to introduce controlled deviations from nominal operating conditions. The phase-angle variation of ±5° was similarly adopted to simulate departures from the ideal 120° phase separation. These ranges are study assumptions, not limits prescribed by the cited standards.

The sampling frequencies of 1, 2, 5, and 10 kHz and observation windows of 2–20 ms were chosen to investigate the trade-off between the quantity of waveform information, estimation accuracy, and observation time. The fifth and seventh harmonics were included because they are relevant components of distorted 50 Hz waveforms, occurring at 250 Hz and 350 Hz, respectively [2], [3]. The harmonic amplitudes, signal-to-noise ratios (SNRs), and DC offsets were varied to test the network under different disturbance severities.

For the combined experiment, 80% of the dataset represents moderate conditions and 20% represents severe conditions. Moderate samples use harmonic amplitudes of 0–5%, SNRs of 30–40 dB, and DC offsets up to 2%; severe samples use harmonic amplitudes of 5–10%, SNRs of 20–30 dB, and DC offsets up to 5%. This mixture is an experimental design choice intended to expose the network to both less-distorted and more challenging signals.

Finally, the dataset is divided into 70% training, 15% validation, and 15% internal testing. These proportions provide data for learning, model selection, and performance evaluation, respectively. The dataset sizes are varied to examine the effect of training-data quantity, while a common independent test set is used where specified to support comparisons between configurations. Overall, the parameter choices establish a controlled simulation framework; validation using measured grid data would be necessary before drawing conclusions about performance in a practical installation.

3.2 Neural Network Architecture and Training

3.2.1 Supervised Learning Framework

The study uses supervised learning, in which the network is trained using input-output pairs. Each input consists of the sampled three-phase voltage waveforms, while its associated output contains the known fundamental amplitudes, phase angles, and frequency.

The network learns a nonlinear mapping of the form

$$\hat{\mathbf{y}} = g\,(\mathbf{x}\,;\,\boldsymbol{\theta})$$

$$\hat{\mathbf{y}} = [V_a,\, V_b,\, V_c,\, \theta_a,\, \theta_b,\, \theta_c, f]^T$$

where ($\mathbf{x}$) is the input vector, ($\hat{\mathbf{y}}$) is the predicted parameter vector,(g) represents the neural network, and($\boldsymbol{\theta}$)denotes its trainable parameters, including weights and biases, (V_a),(V_b),*and*(V_c) represent the voltage amplitudes of phases A, B, and C, respectively; (θ_a),(θ_b),*and*(θ_c)represent their corresponding phase angles; and (f) denotes the fundamental frequency of the three-phase voltage system.

During training, the network predictions are compared with the reference labels. The resulting errors are used to update the trainable parameters repeatedly, allowing the network to learn relationships between the waveform patterns and their corresponding fundamental parameters.

3.2.2 Network Structure and Output Layer

A fully connected feedforward neural network is used for regression. Its hidden layers combine weighted inputs and nonlinear activation functions to learn relationships between the waveform samples and the target parameters. The general structure is:

$$\mathbf{x} \rightarrow \textit{Hiddenlayers} \rightarrow \textit{Outputlayer} \rightarrow \hat{\mathbf{y}}$$

The output layer contains seven neurons, corresponding to the three fundamental amplitudes, three phase angles, and one common fundamental frequency. The network therefore performs a seven-output regression task rather than classifying the waveform into discrete categories.

The implementation used in the dataset-size experiment includes a 36-unit linear bottleneck layer, followed by fully connected layers of 128 and 64 units, with tanh and ReLU activations in the hidden layers and a seven-unit regression output. The other experiments use the architectures specified in their respective scripts.

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3.2.3 Training Procedure and Data Splitting

Each dataset is divided into three subsets:

1. Training set (70%): Used to update the network's trainable parameters.
2. Validation set (15%): Used to monitor training performance and support model selection.
3. Internal test set (15%): Used to evaluate the model on examples not used for training or validation.

The implementations use the Adam optimization algorithm where specified, together with the configured learning rate, mini-batch size, regularization, and learning-rate schedule. Validation-based model selection is used where implemented. The exact training settings vary between experiments and are defined by their respective scripts.

3.3 Experimental Design

Four experiments are conducted to investigate the factors affecting the accuracy of deep-learning-based voltage parameter estimation which is implemented with Matlab.

3.3.1 Experiment 1: Effect of Dataset Size

The first experiment investigates how the quantity of training data influences estimation accuracy. The dataset sizes are 1,000, 2,000, 5,000, 10,000, and 20,000 examples.

The sampling frequency is fixed at 10 kHz and the observation window at 20 ms.. The objective is to determine whether increasing the number of training examples produces consistent improvements in amplitude, phase-angle, and frequency estimation.

3.3.2 Experiment 2: Effect of Sampling Frequency

This experiment evaluates sampling frequencies of 1, 2, 5, and 10 kHz. The observation window is fixed at 20 ms, while dataset sizes are 2,000, 5,000, 10,000, 15,000, and 20,000 examples. The number of samples per phase is 20, 40, 100, and 200, respectively corresponding to 60, 120, 300, and 600 of a 3phase system.

This experiment investigates whether increasing the number of samples within the same observation duration improves estimation accuracy and examines the associated computational implications.

3.3.3 Experiment 3: Effect of Observation Window

This experiment investigates the effect of observation duration on estimation accuracy. The sampling frequency is fixed at 10 kHz, while the observation windows are 2, 5, 10, 15, and 20 ms,while datasets sizes are 2,000, 5,000, 10,000, 15,000, and 20,000 examples.

The experiment therefore evaluates the trade-off between estimation accuracy and observation duration, particularly whether useful estimates can be obtained without waiting for a complete nominal fundamental cycle.

3.3.4 Experiment 4: Combined Sampling Frequency, Observation Window and Datasets Size and

The fourth experiment investigates the combined effects of sampling frequency, observation duration, and datasets size. It uses sampling frequencies of 1, 2, 5, and 10 kHz; observation windows of 5, 10, 15, and 20 ms; and datasets sizes of 10,000 and 20,000 examples. This produce: $4 \times 4 \times 2 = 32$ experimental configurations.

For each configuration, the number of samples per phase is determined by the sampling frequency and observation-window duration. The objective is to identify a practical configuration that balances estimation accuracy, observation time, sampling requirements.

The configuration with the lowest error for one parameter is not necessarily the best overall. The results must be considered across all seven outputs, together with the observation duration and sampling frequency.

3.4 Performance Evaluation Metrics

The network's performance is evaluated by comparing its predictions with the reference values for each test example. The principal error metrics are mean absolute error (MAE) and root mean square error (RMSE). Training time is also recorded to assess computational cost.

Let (y_i) denote a reference value, ($\hat{y}_i$) its predicted value, and (n)the number of test examples.

3.4.1 Mean Absolute Error

MAE measures the average absolute difference between predicted and reference values:

$$\mathrm{MAE}=\frac{1}{n}\sum_{i=1}^{n}|\hat{y}_i-y_i|$$

A lower MAE indicates that the predictions are closer to the reference values on average. The unit depends on the parameter: volts peak for amplitude, degrees for phase angle, and hertz for frequency.

3.4.2 Root Mean Square Error

RMSE is defined as

$$\mathrm{RMSE}=\sqrt{\frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i-y_i)^2}$$

Because the errors are squared before averaging, RMSE gives greater weight to larger errors than MAE. Reporting both metrics provides a more complete assessment of estimation accuracy.

MAE and RMSE are calculated separately for each phase amplitude, each phase angle, and the common frequency. Overall summaries across different parameters must be interpreted carefully because the parameters have different physical units and numerical scales.

3.4.3 Phase-Angle Error

Phase angles are periodic quantities. Direct subtraction can produce an apparent error close to 360° even when two angles are close across the angular boundary.

The wrapped phase-angle error is therefore defined as

$$e_{\phi,i}=\left|mod\left(\hat{\phi}_i-\phi_i+180^\circ\ ,360^\circ\ \right)-180^\circ\ \right|$$

The phase-angle MAE and RMSE are calculated using these wrapped errors. This ensures that the reported error reflects the shortest angular difference between the predicted and reference values.

3.4.4 Training Time and Configuration Comparison

Training time is recorded in seconds for each model. It represents the time required to train the model under the hardware, software, and training settings used; it is not the time required to estimate parameters from a single waveform.

The MAE, RMSE, and training time are compared across configurations. Where common independent test cases are used, they provide a consistent basis for evaluating different models.

The final configuration is assessed using the combined pattern of amplitude, phase-angle, and frequency errors rather than a single metric. Observation duration and sampling frequency are also considered when evaluating practical suitability.

4.EXPERIMENTAL RESULTS AND DISCUSSIONS

4.1 Effect of Dataset Size on Voltage Parameter Estimation (Experiment 1)

This experiment investigated the effect of dataset size on the performance of the neural network in estimating three-phase voltage parameters under disturbances. Five dataset sizes were evaluated: 1,000, 2,000, 5,000, 10,000 and 20,000 examples. The sampling frequency was fixed at 10 kHz, with an observation window of 20 ms and 200 samples per phase.

Table 4.1 Dataset Size on Voltage Parameter Estimation

Dataset size	Training time (s)	Amp A MAE (V)	Amp B MAE (V)	Amp C MAE (V)	Phase A MAE (°)	Phase B MAE (°)	Phase C MAE (°)	Freq MAE (Hz)
1,000	93.7	1.26	1.23	1.2	0.21	0.22	0.2	0.0377
2,000	106.7	1	0.99	0.97	0.22	0.22	0.19	0.0369
5,000	193.8	0.92	0.88	0.82	0.19	0.19	0.17	0.0351
10,000	164.1	1.15	1.23	1.12	0.23	0.24	0.27	0.0402
20,000	175.4	1.37	1.46	1.96	0.23	0.23	0.23	0.0394

From Table 4.1 The 5,000-dataset configuration yields the highest accuracy across almost all metrics:

1. Amplitude MAE: Drops to its lowest point (0.82–0.92 V).
2. Phase MAE: Reaches the best overall accuracy (0.17°–0.19°).
3. Frequency MAE: Achieves the lowest error (0.0351 Hz).

Increasing the dataset from 1,000 to 5,000 provides a clear, quantifiable benefit. Amplitude MAE improves by roughly 30% (from ~1.23 V to ~0.87 V), Phase MAE improves by about 15% (from ~0.21° to ~0.18°), and Frequency MAE improves from 0.0377 Hz to 0.0351 Hz.

Contrary to standard machine learning expectations, expanding the dataset beyond 5,000 in this specific setup causes the model's accuracy to regress:

1. At 10,000, Phase MAE worsens to ~0.24° and Frequency MAE increases to 0.0402 Hz.
2. At 20,000, Amplitude error spikes significantly, with Amp C reaching 1.96 V (more than double the error at 5,000). Phase and Frequency errors remain worse than the 5,000 baseline.

Training time does not scale monotonically with dataset size. While 1,000 datasets train fastest (93.7 s), the 5,000-dataset run actually took the longest (193.8 s). The 10,000 and 20,000 runs were faster (164.1 s and 175.4 s, respectively), suggesting that early stopping or convergence dynamics interact unpredictably with dataset size in this specific architecture.

It is crucial to emphasize that this behavior is not absolute. The optimal dataset size is largely dependent on the complexity and representation of the input data (i.e., the chosen sampling frequency and observation window). As seen in other parts of the broader experiment (such as the combined frequency-window tests), larger datasets (e.g., 20,000) frequently prove more favorable and yield better accuracy when the input representation is different. Therefore, 5,000 is the optimal choice only for this

rigid 10 kHz / 20 ms setup, where the model appears to saturate early and potentially overfit or struggle to find the optimal loss minimum with larger datasets.

4.2 Effect on Frequency Samples on Estimation Performance (Experiment 2)

This experiment investigates how sampling frequency affects the accuracy of neural-network-based estimation of three-phase voltage amplitudes, phase angles, and fundamental frequency. The sampling frequency was varied from 1 kHz to 10 kHz, while the observation window was fixed at 20 ms. Datasets sizes of 2,000, 5,000, 10,000, 15,000, and 20,000 examples were tested to examine whether the effect of sampling frequency depends on the amount of training data. The results were evaluated using mean absolute error (MAE) for each estimated parameter and the model's training time.

Table 4.2.1 Frequency Samples on Estimation Performance

Sampling frequency	Dataset size	Training time (s)	Amp A MAE (V)	Amp B MAE (V)	Amp C MAE (V)	Phase A MAE (°)	Phase B MAE (°)	Phase C MAE (°)	Freq MAE (Hz)
1 kHz	2,000	54.9	1.9	1.76	1.79	0.45	0.49	0.47	0.1
1 kHz	5,000	167.4	1.6	1.69	1.65	0.43	0.47	0.41	0.09
1 kHz	10,000	179.2	1.6	1.65	1.68	0.42	0.46	0.41	0.09
1 kHz	15,000	198.7	1.62	1.7	1.96	0.42	0.52	0.43	0.09
1 kHz	20,000	366.1	1.58	1.69	1.66	0.41	0.46	0.41	0.09
2 kHz	2,000	76.4	1.48	1.44	1.48	0.34	0.38	0.34	0.07
2 kHz	5,000	122.3	1.36	1.33	1.3	0.34	0.38	0.33	0.07
2 kHz	10,000	214	1.31	1.27	1.3	0.33	0.36	0.32	0.07
2 kHz	15,000	366.5	1.24	1.28	1.31	0.32	0.35	0.31	0.07
2 kHz	20,000	413.2	1.25	1.26	1.22	0.33	0.35	0.31	0.07
5 kHz	2,000	78.5	1.17	1.16	1.25	0.25	0.27	0.24	0.05
5 kHz	5,000	136	1.11	1.11	1.16	0.24	0.26	0.23	0.05
5 kHz	10,000	242.3	1.09	1	1	0.24	0.25	0.22	0.04
5 kHz	15,000	175.7	1.25	1.23	1.46	0.25	0.29	0.24	0.05
5 kHz	20,000	416.2	**0.92**	0.89	0.95	0.22	0.25	0.22	0.04
10 kHz	2,000	86.4	0.95	0.96	1.03	0.2	0.22	0.18	0.04
10 kHz	5,000	64.6	1.64	1.61	1.78	0.24	0.25	0.21	0.04
10 kHz	10,000	164.9	1.09	1.13	1.22	0.22	0.25	0.2	0.04
10 kHz	15,000	221.8	0.95	1.08	1.08	0.21	0.29	0.21	0.05
10 kHz	20,000	264	1.07	1	1.23	0.21	0.23	0.25	0.04

Table 4.2.2 Result Summary

Sampling Frequency	Dataset Size	Mean Amp MAE (V)	Mean Phase MAE (°)	Freq MAE (Hz)	Training Time (s)
1 kHz	2000	1.82	0.47	0.1	54.9
1 kHz	20000	1.64	0.43	0.09	366.1
2 kHz	20000	1.24	0.33	0.07	413.2
5 kHz	20000	0.92	0.23	0.04	416.2
10 kHz	2000	0.98	0.2	0.04	86.4
10 kHz	20000	1.1	0.23	0.04	264
5 kHz (Best Compromise)	5000	1.13	0.24	0.05	136
5 kHz (Best Compromise)	10000	1.03	0.24	0.04	242.3

At 1 kHz, with a fixed 20 ms observation window, the mean amplitude MAE at a dataset size of 20,000 is 1.64 V, while the mean phase MAE is 0.43° and frequency MAE is 0.09 Hz. Increasing the dataset size from 2,000 to 20,000 reduces the mean amplitude MAE from approximately 1.81 V to 1.64 V, but the improvement in phase and frequency estimation is relatively small.

At 2 kHz, estimation accuracy improves compared with 1 kHz. For 20,000 examples, the mean amplitude MAE decreases to 1.24 V, the mean phase MAE to 0.33°, and frequency MAE to 0.07 Hz. Increasing the dataset size generally improves amplitude estimation, although phase and frequency errors change only slightly.

At 5 kHz, performance improves further. With 20,000 examples, the mean amplitude MAE falls to 0.92 V, the mean phase MAE to 0.23°, and frequency MAE to 0.04 Hz. However, increasing the dataset from 2,000 to 10,000 examples produces only modest improvements in phase and frequency estimation, while training time increases from 78.5 s to 242.3 s.

10 kHz frequency sample generally provides the lowest phase and frequency errors. With 2,000 examples, the mean phase MAE is approximately 0.20° and frequency MAE is 0.04 Hz, with a training time of 86.4 s. Increasing the dataset to 20,000 examples does not consistently improve accuracy: the mean phase MAE is approximately 0.23°, frequency MAE remains 0.04 Hz, and training time rises to 264.0 s.

Overall, Increasing the sampling frequency from 1 kHz to 10 kHz generally improves estimation accuracy, particularly for phase and frequency. However, the gains become smaller at higher sampling frequencies, and increasing the dataset size does not guarantee further improvement. A sampling frequency of 5 kHz is a promising compromise: with 5,000–10,000 examples, it achieves a mean phase MAE of approximately 0.24° and a frequency MAE of 0.04 – 0.05 Hz, with training times of 136.0 – 242.3 s. It also requires only 300 input features for a 20 ms window (50 samples per phase). Although 10 kHz can provide slightly better phase and frequency estimates, 5 kHz offers a practical balance between accuracy, input size, and training cost.

4.3 Effect Observation Window on Estimation Performance (Experiment 3)

This experiment investigated the effect of observation window duration on neural network estimation performance. The sampling frequency was fixed at 10 kHz, while the observation window was varied across 2, 5, 10, 15 and 20 ms. Five dataset sizes, ranging from 2,000 to 20,000 examples, were evaluated. The model estimated the three phase-voltage peak amplitudes, phase angles and shared fundamental frequency. Performance was assessed using mean absolute error (MAE), root mean square error (RMSE) and training time. The results are presented in Table 4.4.1.

Table 4.3.1: Effect of Observation Window and Dataset Size on Estimation Performance

Window (ms)	Dataset size	Training (s)	Amp A MAE (V)	Amp B MAE (V)	Amp C MAE (V)	Phase A MAE (°)	Phase B MAE (°)	Phase C MAE (°)	Freq MAE (Hz)
2	2,000	84.3	12.77	15.39	14.34	2.49	2.32	2.48	0.26
2	5,000	188.2	12.5	15.34	14.27	2.48	2.31	2.49	0.26
2	10,000	266.3	12.37	15.34	14.19	2.48	2.32	2.48	0.26
2	15,000	508.3	12.41	15.36	14.21	2.48	2.31	2.47	0.26
2	20,000	560.5	12.37	15.37	14.15	2.48	2.31	2.47	0.26
5	2,000	58.6	9.56	13.82	4.47	1.76	0.87	2.37	0.25
5	5,000	115.2	9.44	13.66	4.25	1.74	0.86	2.4	0.25
5	10,000	455.3	9.3	13.65	4.2	1.72	0.86	2.34	0.25
5	15,000	731.4	9.26	13.55	4.16	1.72	0.86	2.32	0.25

5	20,000	1,027.30	9.24	13.6	4.14	1.72	0.86	2.29	0.25
10	2,000	106.4	3.12	1.82	1.88	0.45	0.44	0.67	0.23
10	5,000	71.9	3.28	2.24	2.35	0.46	0.44	0.67	0.24
10	10,000	427.9	2.99	1.67	1.67	0.45	0.41	0.63	0.23
10	15,000	257.9	3.12	1.77	1.8	0.48	0.41	0.65	0.23
10	20,000	378.6	3.12	1.77	1.74	0.46	0.42	0.67	0.23
15	2,000	108.6	1.26	1.08	1.36	0.26	0.34	0.33	0.09
15	5,000	280.3	1.01	0.9	1.15	0.24	0.32	0.32	0.09
15	10,000	199.8	1.74	1.22	2.07	0.47	0.65	0.69	0.22
15	15,000	297.6	1.38	2.51	1.4	0.25	0.32	0.32	0.08
15	20,000	399	2.32	1.49	1.48	0.25	0.4	0.37	0.09
20	2,000	91.9	1.04	1.03	1.08	0.2	0.22	0.2	0.04
20	5,000	114.8	1.14	1.23	1.32	0.25	0.25	0.27	0.05
20	10,000	272.3	0.92	0.98	0.87	0.2	0.21	0.18	0.04
20	15,000	241.4	1.15	1.26	1.2	0.23	0.26	0.31	0.04
20	20,000	319.2	1.38	1.49	1.68	0.23	0.26	0.28	0.04

Table 4.3.2 Result Summary

Window (ms)	Dataset Size	Mean Amp MAE (V)	Mean Phase MAE (°)	Freq MAE (Hz)	Training Time (s)
2 ms	20000	13.96	2.42	0.26	560.5
5 ms	20000	9	1.62	0.25	1027.3
10 ms	10000	2.11	0.5	0.23	427.9
15 ms	5000	1.02	0.29	0.09	280.3
20 ms	20000	1.52	0.26	0.04	319.2
20 ms	2000	105	0.21	0.04	91.9
20 ms (Strongest Overall)	10000	0.92	0.2	0.04	272.3
15 ms (Faster Compromise)	5000	1.02	0.29	0.09	280.3

At 2 ms: Estimation accuracy is poor across all dataset sizes. With 20,000 examples, the mean amplitude MAE remains approximately 13.96 V, the mean phase MAE is 2.42°, and the frequency MAE is 0.26 Hz. Increasing the dataset from 2,000 to 20,000 examples produces little improvement despite training time increasing from 84.3 s to 560.5 s. This suggests that the short observation window, rather than dataset size alone, limits estimation performance.

At 5 ms: Accuracy improves compared with 2 ms, but substantial errors remain. For 20,000 examples, the mean amplitude MAE is approximately 9.00 V, the mean phase MAE is 1.62°, and the frequency MAE is 0.25 Hz. Increasing the dataset size produces only modest improvements, while training time rises to 1,027.3 s. Thus, adding more training data does not compensate for the limited observation window.

At 10 ms: Estimation accuracy improves considerably. With 10,000 examples, the mean amplitude MAE is approximately 2.11 V, the mean phase MAE is 0.50°, and the frequency MAE is 0.23 Hz. However, increasing the dataset size does not consistently reduce errors, and frequency estimation remains less accurate than with longer windows.

At 15 ms: Amplitude and phase estimation improve further. With 5,000 examples, the mean amplitude MAE is approximately 1.02 V, the mean phase MAE is 0.29°, and the frequency MAE is 0.09 Hz. However, results vary across dataset sizes. In particular, the 10,000-example configuration produces

noticeably larger errors, including a frequency MAE of 0.22 Hz. Therefore, increasing the dataset size does not guarantee better performance.

At 20 ms: This window generally provides the best overall estimation accuracy. With 10,000 examples, the mean amplitude MAE is approximately 0.92 V, the mean phase MAE is 0.20°, and the frequency MAE is 0.04 Hz. Increasing the dataset to 20,000 examples does not improve the results; the mean amplitude MAE rises to approximately 1.52 V, while phase and frequency errors remain similar. The 2,000-example configuration also performs reasonably well, achieving a mean amplitude MAE of 1.05 V, mean phase MAE of 0.21°, and frequency MAE of 0.04 Hz.

Overall, increasing the observation window from 2 ms to 20 ms substantially improves amplitude, phase, and frequency estimation. The improvement is particularly pronounced between 5 ms and 15 ms, while the 20 ms window achieves the lowest errors overall. Dataset size has a less consistent effect: larger datasets sometimes improve accuracy, but can also increase training time without meaningful gains.

4.4 Combined Effect of Sampling Frequency and Observation Window on Estimation Performance

The final experiment investigated the combined effects of sampling frequency, observation window and dataset size on voltage parameter estimation accuracy. Four sampling frequencies (1, 2, 5 and 10 kHz), five observation windows (5, 10, 15 and 20 ms), and two dataset sizes (10,000 and 20,000 examples) were evaluated, giving 40 configurations per training seed.

Table 4.4.1 Combined Effect of Sampling Frequency and Observation Window on Estimation Performance

Sampling frequency (kHz)	Window (ms)	Dataset size	Training time (s)	Amp A MAE (V)	Amp B MAE (V)	Amp C MAE (V)	Phase A MAE (°)	Phase B MAE (°)	Phase C MAE (°)	Freq MAE (Hz)
1	5	10000	313.6	5.78	3.85	6.27	0.86	1.37	0.73	0.25
1	5	20000	319.4	5.8	3.84	6.22	0.85	1.37	0.73	0.25
2	5	10000	357.3	4.38	3.28	4.42	0.74	1.08	0.59	0.25
2	5	20000	657.4	4.32	3.19	4.41	0.74	1.05	0.58	0.25
5	5	10000	104	3.44	2.96	3.42	0.64	0.87	0.54	0.25
5	5	20000	530.7	3.25	2.82	3.21	0.59	0.81	0.52	0.25
10	5	10000	340.8	2.76	2.57	2.58	0.5	0.68	0.51	0.25
10	5	20000	373.4	2.81	2.57	2.54	0.52	0.69	0.5	0.25
1	10	10000	324.9	3.24	2.65	2.47	0.51	0.67	0.63	0.2
1	10	20000	672.3	3.17	2.6	2.44	0.51	0.64	0.62	0.19
2	10	10000	334.8	2.8	2.14	1.99	0.4	0.54	0.54	0.17
2	10	20000	665.2	2.85	2.13	2.04	0.4	0.54	0.54	0.17
5	10	10000	325.4	2.26	1.59	1.57	0.33	0.43	0.45	0.15
5	10	20000	400.8	2.31	1.55	1.63	0.33	0.44	0.46	0.15
10	10	10000	190	2.17	1.43	1.39	0.32	0.38	0.43	0.14
10	10	20000	425.9	2.13	1.34	1.38	0.33	0.38	0.42	0.14
1	15	10000	318.4	2.13	1.84	2.27	0.44	0.56	0.5	0.14
1	15	20000	680	2.09	1.82	2.25	0.44	0.55	0.49	0.13
2	15	10000	355.5	1.52	1.37	1.68	0.37	0.49	0.42	0.12
2	15	20000	687.2	1.56	1.35	1.66	0.35	0.48	0.42	0.12

5	15	10000	356	1.1	0.94	1.3	0.29	0.37	0.34	0.1
5	15	20000	409.4	1.2	0.98	1.33	0.27	0.37	0.33	0.09
10	15	10000	373.7	0.89	0.81	1.03	0.25	0.32	0.29	0.08
10	15	20000	496.9	0.9	0.77	1.01	0.25	0.33	0.3	0.08
1	20	10000	188.5	1.63	1.66	1.64	0.44	0.46	0.42	0.09
1	20	20000	314.7	1.58	1.59	1.67	0.43	0.46	0.41	0.09
2	20	10000	174.7	1.27	1.28	1.27	0.34	0.37	0.33	0.07
2	20	20000	672.2	1.23	1.18	1.2	0.31	0.35	0.31	0.06
5	20	10000	166.6	0.97	0.97	1.01	0.25	0.27	0.24	0.05
5	20	20000	433.5	0.94	0.93	0.91	0.23	0.27	0.24	0.04
10	20	10000	234	0.82	0.78	0.87	0.21	0.23	0.2	0.04
10	20	20000	363.2	0.7	0.72	0.8	0.21	0.23	0.2	0.04

(Refer to appendix A for full RSME table)

Table 4.4.2 Result Summary

fs (kHz)	Window (ms)	Dataset	Amp MAE (V)	Amp RMSE (V)	Phase MAE (°)	Phase RMSE (°)	Freq MAE (Hz)	Freq RMSE (Hz)	Train (s)
5	5	20k	3.09	4.24	0.64	0.86	0.25	0.29	530.7
5	20	20k	0.93	1.27	0.25	0.34	0.04	0.06	433.5
5	20	10k	0.98	1.35	0.25	0.35	0.05	0.07	166.6
10	20	10k	0.83	1.1	0.21	0.29	0.04	0.05	234
5	20	10k	0.98	1.35	0.25	0.35	0.05	0.07	166.6
5	20	20k	0.93	1.27	0.25	0.34	0.04	0.06	433.5
5	20	10k	0.98	1.35	0.25	0.35	0.05	0.07	166.6
10	20	20k	0.74	1.01	0.21	0.29	0.04	0.05	363.2
5	10	10k	1.81	2.49	0.4	0.54	0.15	0.19	325.4

4.4.1 The "Goldilocks" Analysis: Balancing Sampling and Window Size

1. General Trends: The Window Dominates

The observation window length has a substantial impact on estimation accuracy. At a fixed sampling frequency of 5 kHz, increasing the observation window from 5 ms to 20 ms reduces the Phase MAE from approximately 0.64° to 0.25° and the Frequency MAE from approximately 0.25 Hz to 0.05 Hz. The corresponding mean phase RMSE decreases from approximately 0.86° to 0.34°, while frequency RMSE decreases from approximately 0.29 Hz to 0.07 Hz.

At 50 Hz, a 20 ms observation window captures one complete fundamental cycle, providing more waveform information for estimating phase and frequency. Shorter windows generally produce larger estimation errors, particularly at 2 ms and 5 ms. However, the results indicate that accuracy improves progressively with window length rather than establishing a universal minimum window of 15 ms.

2. Sampling Frequency and Diminishing Returns

Increasing the sampling frequency generally improves amplitude and phase estimation at a fixed observation window. However, the improvement becomes less pronounced at higher sampling frequencies, indicating diminishing returns in relation to the additional input data.

At a 20 ms observation window, the following results were obtained:

1. 5 kHz, 10,000 training examples: Amplitude MAE = 0.98 V; amplitude RMSE = 1.35 V; phase MAE = 0.25°; phase RMSE = 0.35°; frequency MAE = 0.05 Hz; frequency RMSE = 0.07 Hz.
2. 10 kHz, 10,000 training examples: Amplitude MAE = 0.83 V; amplitude RMSE = 1.10 V; phase MAE = 0.21°; phase RMSE = 0.29°; frequency MAE = 0.04 Hz; frequency RMSE = 0.05 Hz.

The 10 kHz configuration achieves lower errors, but it doubles the input dimension from 300 to 600 features compared with 5 kHz. The improvement must therefore be considered alongside the additional sampling and input-processing requirements. RMSE is consistently higher than MAE, indicating that larger prediction errors contribute more strongly to the squared-error metric.

3. Dataset Size Versus Training Time

Increasing the dataset from 10,000 to 20,000 examples does not guarantee a proportional improvement in accuracy. For the 5 kHz, 20 ms configuration, amplitude MAE decreases from 0.98 V to 0.93 V, while amplitude RMSE decreases from 1.35 V to 1.27 V. Phase MAE improves from 0.25° to 0.25°, and phase RMSE from 0.35° to 0.34°. Frequency MAE decreases from 0.05 Hz to 0.04 Hz, while frequency RMSE decreases from 0.07 Hz to 0.06 Hz.

However, training time increases from 166.60 seconds to 433.50 seconds. This suggests that the additional training cost may outweigh the relatively modest accuracy improvements for some applications. Nevertheless, these results alone do not establish that 10,000 examples are universally sufficient for convergence; they indicate that the benefit of increasing the dataset depends on the configuration and the accuracy requirements.

4.4.2 Choosing the Best Configuration Based on Application

Different applications may prioritise estimation accuracy, observation time, input size, or training cost. Based on the combined experiment, three configurations illustrate these trade-offs.

1. The "Absolute Precision" Candidate (For Offline Processing)

Configuration: 10 kHz | 20 ms | 20,000 training examples

Performance
Mean amplitude MAE: 0.74 V; mean amplitude RMSE: 1.01 V.
Mean phase MAE: 0.21°; mean phase RMSE: 0.29°.
Frequency MAE: 0.04 Hz; frequency RMSE: 0.05 Hz.
Training time: 363.20 seconds.

This configuration achieved the lowest overall error among the tested configurations in the reported run. It is a promising candidate for offline grid analysis and applications prioritising estimation accuracy. However, suitability for protection relays or other real-time systems cannot be established from these results alone, because inference latency and hardware performance were not evaluated.

2. The "Optimal Balance" Candidate (For Monitoring Applications)

Configuration: 5 kHz | 20 ms | 10,000 training examples

Performance:
Mean amplitude MAE: 0.98 V; mean amplitude RMSE: 1.35 V.
Mean phase MAE: 0.25°; mean phase RMSE: 0.35°.
Frequency MAE: 0.05 Hz; frequency RMSE: 0.07 Hz.
Training time: 166.60 seconds.

This configuration represents a potential compromise between estimation accuracy and sampling requirements. Compared with 10 kHz, it halves the sampling frequency and reduces the input dimension from 600 to 300 features while retaining a complete 20 ms fundamental-cycle observation window. Its errors are somewhat higher than those of the 10 kHz configuration, but it may be suitable where these error levels meet the application requirements. Its shorter training time does not, by itself, demonstrate faster real-time inference or suitability for microcontroller deployment.

3. The "Low-Latency" Candidate (For Shorter Observation Windows)

Configuration: 5 kHz | 10 ms | 10,000 training examples

Performance:
Mean amplitude MAE: 1.81 V; mean amplitude RMSE: 2.49 V.
Mean phase MAE: 0.40°; mean phase RMSE: 0.54°.
Frequency MAE: 0.15 Hz; frequency RMSE: 0.19 Hz.
Training time: 325.40 seconds.

This configuration reduces the observation window from 20 ms to 10 ms, potentially allowing more frequent parameter updates. At 50 Hz, however, a 10 ms window captures only half a fundamental cycle. The results show higher amplitude, phase, and frequency errors than the 5 kHz, 20 ms configuration. It may therefore be worth considering when shorter observation time is important and the resulting estimation errors are acceptable. Actual dynamic tracking performance would require separate evaluation.

Overall, the combined experiment identifies 5 kHz sampling, a 20 ms observation window, and 10,000 training examples as a potential compromise between estimation accuracy and reduced sampling requirements. The 10 kHz, 20 ms configuration achieves better accuracy, while the 10 ms configuration offers a shorter observation window at the cost of increased errors. MAE provides an interpretable measure of average error, while RMSE complements it by highlighting the influence of larger errors. The final configuration should therefore be selected according to the required accuracy and observation time, with further testing needed before claiming suitability for real-time deployment.

CHAPTER FIVE: CONCLUSION

5.1 Conclusion

This study investigated the effects of dataset size, sampling frequency, and observation window on deep learning-based estimation of three-phase voltage amplitudes, phase angles, and fundamental frequency under disturbances.

The results showed that longer observation windows and higher sampling frequencies generally improved estimation accuracy, while increasing dataset size did not guarantee further improvement. The 10 kHz, 20 ms, 10,000-example configuration achieved mean amplitude MAE and RMSE of 0.83 V and 1.10 V, phase-angle MAE and RMSE of 0.21° and 0.29°, and frequency MAE and RMSE of 0.04 Hz and 0.05 Hz, respectively. Increasing the dataset to 20,000 examples reduced amplitude MAE to 0.74 V and RMSE to 1.01 V, but increased training time from approximately 234 to 363 seconds.

The 5 kHz, 20 ms, 10,000-example configuration offered a potential compromise, achieving amplitude MAE and RMSE of 0.98 V and 1.35 V, phase-angle MAE and RMSE of 0.25° and 0.35°, and frequency MAE and RMSE of 0.05 Hz and 0.07 Hz. While MAE measures average prediction error, RMSE gives greater weight to larger errors.

Overall, the 5 kHz configuration offers a potential balance between accuracy and reduced sampling requirements, while 10 kHz provides better accuracy. Further validation using multiple experimental runs, real voltage data, and inference-latency testing is necessary before practical deployment.

5.2 Limitations and Future Work

The study relied on simulated voltage signals and a selected neural network architecture. Therefore, the findings may not fully represent real power-grid conditions. Future work should validate the model using measured voltage data, compare it with conventional estimation methods such as least-squares fitting and Fourier-based estimation, and investigate alternative neural network architectures.

Further research should also examine shorter observation windows, particularly where rapid voltage parameter updates are required. In addition, inference latency and computational requirements on target hardware should be measured, since training time alone does not establish real-time capability. Finally, future work could develop an adaptive approach that selects the sampling frequency, observation window, and dataset size according to the required estimation accuracy and available computational resources.

References

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APPENDIX

APPENDIX A: Goldilocks Experiment — Complete RMSE Results

Table A1: Complete Results for Combined Sampling Frequency, Observation Window, and Dataset Size Experiment.

fs (kHz)	Window (ms)	Dataset	Amp A RMSE (V)	Amp B RMSE (V)	Amp C RMSE (V)	Phase A RMSE (º)	Phase B RMSE (º)	Phase C RMSE (º)	Freq RMSE (Hz)
1	5	10k	7.77	5.29	8.26	1.17	1.76	0.99	0.29
1	5	20k	7.77	5.29	8.17	1.16	1.76	0.98	0.29
2	5	10k	5.83	4.5	6.01	1.03	1.46	0.78	0.29
2	5	20k	5.8	4.39	5.97	1.02	1.43	0.78	0.29
5	5	10k	4.64	4.03	4.67	0.88	1.17	0.71	0.29
5	5	20k	4.41	3.88	4.44	0.82	1.09	0.68	0.29
10	5	10k	3.73	3.49	3.45	0.7	0.94	0.67	0.29
10	5	20k	3.82	3.51	3.41	0.72	0.94	0.66	0.29
1	10	10k	4.51	3.73	3.57	0.7	0.92	0.85	0.24
1	10	20k	4.39	3.69	3.51	0.69	0.89	0.83	0.23
2	10	10k	3.83	2.94	2.78	0.55	0.73	0.73	0.22
2	10	20k	3.84	2.91	2.81	0.55	0.73	0.72	0.21
5	10	10k	3.14	2.18	2.15	0.43	0.58	0.6	0.19
5	10	20k	3.19	2.13	2.19	0.44	0.59	0.6	0.19
10	10	10k	3	1.9	1.89	0.41	0.51	0.58	0.18
10	10	20k	2.95	1.79	1.88	0.42	0.51	0.57	0.18
1	15	10k	3.05	2.72	3.3	0.61	0.76	0.65	0.18
1	15	20k	3	2.67	3.28	0.6	0.74	0.64	0.17
2	15	10k	2.17	2.02	2.44	0.5	0.67	0.56	0.16
2	15	20k	2.22	2	2.44	0.48	0.67	0.55	0.15
5	15	10k	1.55	1.31	1.77	0.38	0.51	0.45	0.13
5	15	20k	1.63	1.34	1.78	0.36	0.5	0.43	0.12
10	15	10k	1.22	1.09	1.38	0.34	0.43	0.4	0.11
10	15	20k	1.22	1.08	1.36	0.34	0.44	0.4	0.11
1	20	10k	2.36	2.48	2.34	0.6	0.63	0.58	0.12
1	20	20k	2.3	2.42	2.36	0.59	0.64	0.57	0.12
2	20	10k	1.83	1.89	1.83	0.47	0.51	0.47	0.09
2	20	20k	1.77	1.78	1.75	0.43	0.48	0.44	0.09
5	20	10k	1.34	1.32	1.39	0.35	0.37	0.32	0.07
5	20	20k	1.29	1.26	1.28	0.32	0.37	0.32	0.06
10	20	10k	1.08	1.07	1.16	0.29	0.31	0.27	0.05
10	20	20k	0.95	0.99	1.09	0.28	0.31	0.27	0.05

APPENDIX B: Goldilocks Experiment — Prediction Accuracy Visualisations

Figure B1: True Versus Predicted Parameters on the Common Test Set (10 kHz, 20,000 Training Examples).

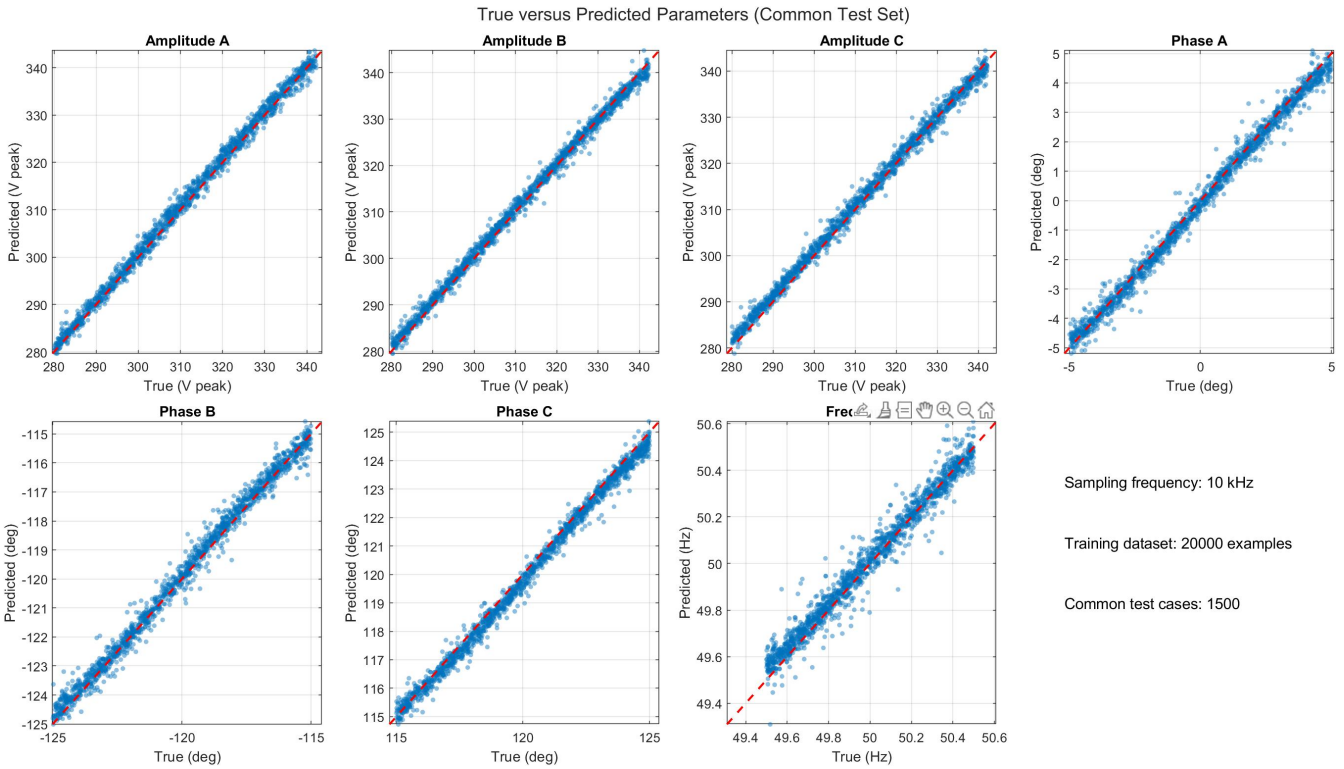


Figure B2: Common-Test RMSE Across Sampling Frequencies and Observation Windows (Dataset Size = 20,000).

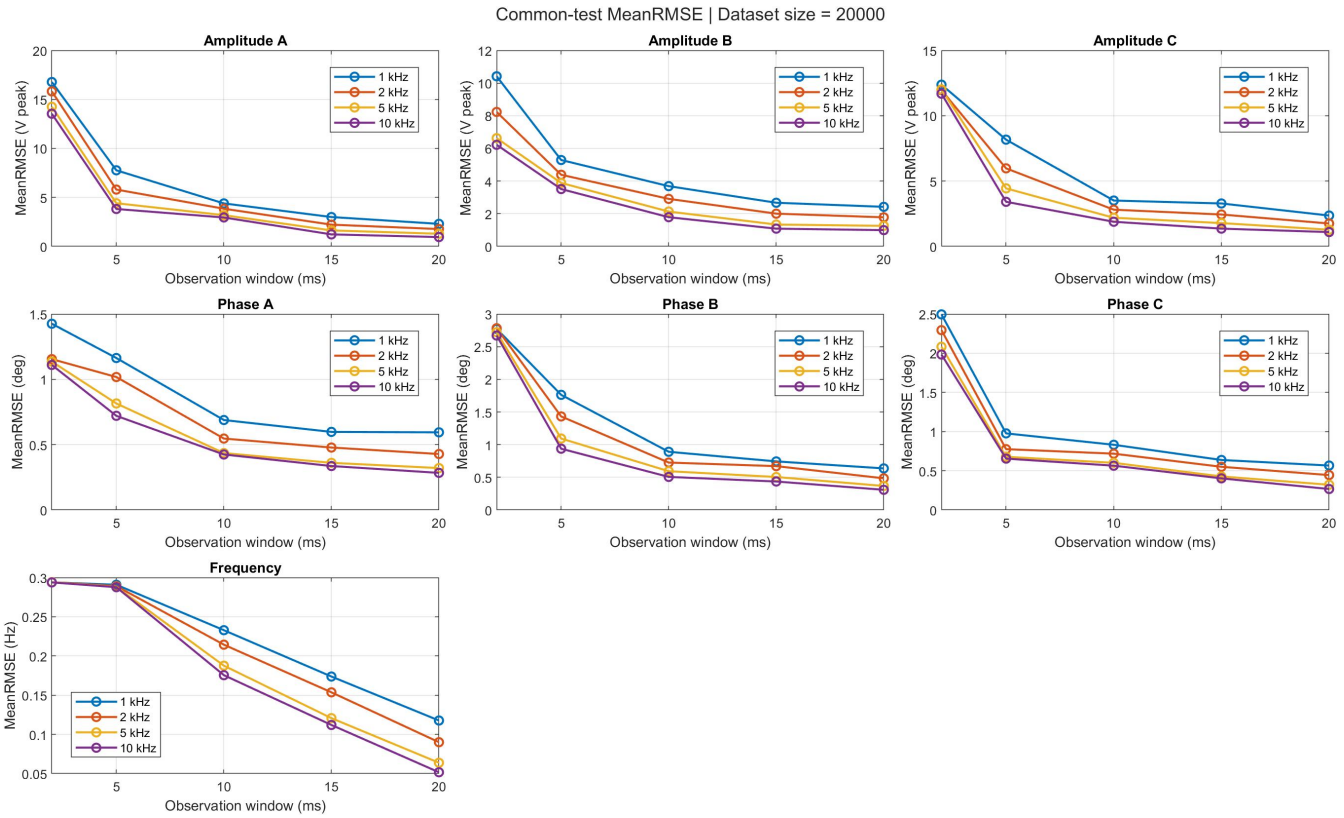


Figure B3: Common-Test MAE Across Sampling Frequencies and Observation Windows (Dataset Size = 10,000).

